CS229-Probability Review

1-Elements of Probability

1.1-Defination-Axioms of Probability

- Sample space Ω: The set of all the outcomes of a random experiment.
- Event space $\mathcal{F}$: A set whose elements $A \in \mathcal{F}$ are subsets of Ω.
- Probability measure: A function $P : \mathcal{F} \to \mathbb{R}$ that satisfying the following properties:
  - **Non-negativity**: $P(A) \geq 0$, for all $A \in \mathcal{F}$.
  - **Completeness**: $P(\Omega) = 1$.
  - Countable Additivity: if $A_1, A_2, \cdots$ are **disjoint events** (i.e. $A_i \cap A_j = \emptyset$ whenever $i \neq j$), then

$$P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$$

1.2-Properties of probability

- If $A \subseteq B$, then $P(A) \leq P(B)$.
- $P(A \cap B) \leq min(P(A), P(B))$.
- $P(\complement_{\Omega} A) := P(\Omega \setminus A) = 1 - P(A)$.
- $P(A \cup B) \leq P(A) + P(B)$.
- If $A_1, \cdots, A_k$ are a set of **disjoint events** such that $\bigcup_{i=1}^{k} A_i = \Omega$, then $\sum_{i=1}^{k} P(A_k) = 1$ (Easy to prove by Countable Additivity).

1.3-Conditional probability, independence and Beyes' theorem

- The conditional probability of any event A given B is defined as

$$P(A|B) := \frac{P(A \cap B)}{P(B)}$$

- Mutually independent:

$$P(\bigcap_{i \in S} A_i) = \prod_{i \in S} P(A_i), \forall S \subseteq \{1, 2, \cdots, k\}$$

Law of total probability

Suppose $A_i, \cdots, A_n$ are disjoint events, and event B satisfies $B \subseteq \bigcup_{i=1}^n A_i$, then

$$P(B) = \sum_{i=1}^{n} P(A_i)P(B|A_i)$$

And an important corollary of the law of total probability is Beyes' theorem

Beyes' theorem

Suppose $A_1, \cdots, A_n$ are disjoint events, and event B satisfies $B \subset \bigcup_{i=1}^n A_i$. Then if $P(B) > 0$, it is the case that

$$P(A_j|B) = \frac{P(A_j)P(B|A_j)}{\sum_{i=1}^n P(A_i)P(B|A_i)}$$

Proof.
By the fact that

$$P(A_j \cap B) = P(A_j)P(B|A_j) = P(B)P(A_j|B)$$

Then,

$$P(A_j|B) = \frac{P(A_j)P(B|A_j)}{P(B)}$$

Because of the law of total probability, the Beyes' theorem is proved.

2-Random Variables

2.1-Definition

- A random variable X is a function $X : \Omega \to \mathbb{R}$.
- If $X(\omega)$ can take only a finite number of values, it is known as a **discrete random variable**.
- The probability of the set associated with a random variable X taking on some specific value k is

$$P(X = k) := P(\{\omega : X(\omega) = k\})$$

- If $X(\omega)$ takes on a infinite number of possible values, it is called a **continuous random variable**.
- The probability that X takes on a value between two real constants is denoted as

$$P(a \leq X \leq b) := P(\{\omega : a \leq X(\omega) \leq b\})$$

2.2-Cumulative distribution functions

A **cumulative distribution function** (CDF) is a function $F_X : \mathbb{R} \to [0, 1]$ which specifies a probability measure as

$$F_X(x) := P(X \leq x)$$

A CDF function satisfies the following properties.

- $0 \leq F_X(x) \leq 1$
- $\lim_{x \to -\infty} F_X(x) = 0$
- $\lim_{x \to \infty} F_X(x) = 1$
- $x \leq y \to F_X(x) \leq F_X(y)$

2.3-Probability mass functions

A **probability mass function** (PMF) is a function $p_X : \Omega \to \mathbb{R}$ such that

$$p_X(x) := P(X = x)$$

Notation.
$Val(X)$ represents the set of possible values that the random variable X may assume.

A PMF function satisfies the following properties.

- $0 \leq p_X(x) \leq 1$
- $\sum_{x \in Val(X)} p_X(x) = 1$
- $\sum_{x \in A} p_X(x) = P(X \in A)$

2.4-Probability density functions

> **Important**
>
> The cumulative distribution function $F_X(x)$ is differentiable everywhere for some continuous random variables

A **Probability Density Function** (PDF) is defined by

$$f_X(x) := \frac{\mathrm{d}F}{\mathrm{d}x}$$

So, $P(x \leq X \leq x + \Delta x) = \Delta F = f_X(x)\Delta x + o(\Delta x) \approx f_X(x)\Delta x$

A PDF function satisfies the following properties.

- $f_X(x) \geq 0$
- $\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = 1$

- $\int_{x \in A} f_X(x)\mathrm{d}x = P(X \in A)$

2.5-Expectation

If X is a discrete random variable with PMF $p_X(x)$ and $g : \mathbb{R} \to \mathbb{R}$ is an arbitary function. In this case, $g(X)$ can be considered a random variable, and we define the **expectation** or **expected value** of $g(X)$ as

$$E[g(X)] := \sum_{x \in Val(X)} g(x)p_X(x)$$

if X is a continuous random variable with PDF $f_X(x)$, then the expected value of $g(X)$ is defined as

$$E[g(X)] := \int_{-\infty}^{\infty} g(x)f_X(x)\mathrm{d}x$$

The **mean** of the random variable X is

$$E[X] = \int_{-\infty}^{\infty} xf_X(x)\mathrm{d}x$$

Expectation satisfies the following properties.

- $E[a] = a$ for any constant $a \in \mathbb{R}$
- $E[af(X)]aE[f(X)]$ for any constant $a \in \mathbb{R}$
- $E[f(X) + g(X)] = E[f(X)] + E[g(X)]$. This property is known as the **linearity of expectation**
- For a discrete random variable X, $E[1\{X = k\}] = P(X = k)$

2.6-Variance

The variance of a random variable X is a measure of how concentrated the distribution of a random variable X is around its mean. Formally, the **variance** of a random variable X is defined as

$$Var[X] := E[(X - E[X])^2]$$

And then, we can calculate

$$\begin{aligned} E[(X - E[X])^2] &= E[X^2 - 2E[X]X + E[X]^2] \\ &= E[X^2] - 2E[X]E[X] + E[X]^2 \\ &= E[X^2] - E[X]^2 \end{aligned}$$

Variance satisfies the following properties.

- $Var[a] = 0$ for any constant $a \in \mathbb{R}$
- $Var[af(X)] = a^2Var[f(X)]$ for any constant $a \in \mathbb{R}$

2.7-Some common distributions

Discrete random variables

- $X \sim Bernoulli(p)$ (where $0 \leq p \leq 1$): one if a coin with heads probability p comes up heads, zero otherwise.

$$p(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

- $X \sim Binomial(n, p)$ (where $0 \leq p \leq 1$): the number of heads in n independent flips of a coin with heads probability p.

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

- $X \sim Geometric(p)$ (where $p > 0$): the number of flips of a coin with heads probability p until the first heads.

$$p(x) = p(1-p)^{x-1}$$

- $X \sim Poisson(\lambda)$ (where $\lambda > 0$): a probability distribution over the nonnegative integers used for modeling the frequency of rare events.

$$p(x) = e^{-\lambda} \frac{\lambda^x}{x!}$$

Continuous random variables

- $X \sim Uniform(a, b)$ (where $a < b$): equal probability density to every value between a and b on the real line.

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- $X \sim Exponential(\lambda)$ (where $\lambda > 0$): decaying probability density over the nonnegative reals.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- $X \sim Normal(\mu, \sigma^2)$: also known as the Gaussian distribution.

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$DIstribution$	$Mean$	$Variance$
$Bernoulli(p)$	p	$p(1-p)$

$DIstribution$	$Mean$	$Variance$
$Binomial(n,p)$	np	$np(1-p)$
$Geometric(p)$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
$Poisson(\lambda)$	λ	λ
$Uniform(a,b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$Gaussian(\mu,\sigma^2)$	μ	σ^2
$Exponential(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

3-Two Random Variables

3.1-Joint and marginal distributions

A **joint cumulative distribution function** of X and Y is defined by

$$F_{XY}(x,y) := P(X \leq x, Y \leq y)$$

The joint CDF $F_{XY}(x,y)$ and the joint distribution functions $F_X(x)$ and $F_Y(y)$ of each variable separately are related by

$$F_X(x) = \lim_{y\to\infty} F_{XY}(x,y)$$
$$F_Y(y) = \lim_{x\to\infty} F_{XY}(x,y)$$

Here, we call $F_X(x)$ and $F_Y(y)$ the **marginal cumulative distribution functions** of $F_{XY}(x,y)$.

The joint CDF satisfies the following properties.

- $0 \leq F_{XY}(x,y) \leq 1$
- $\lim_{x,y\to\infty} F_{XY}(x,y)$
- $\lim_{x,y\to-\infty} F_{XY}(x,y) = 0$
- $F_X(x) = \lim_{y\to\infty} F_{XY}(x,y)$

3.2-Joint and marginal probability mass functions

Let X and Y be discrete random variables, then the **joint probability mass function** $p_{XY} : \mathbb{R}\times\mathbb{R} \to [0,1]$ is defined by

$$p_{XY}(x,y) := P(X = x, Y = y)$$

Then the **marginal probability mass function** is defined by

$$p_X(x) = \sum_y p_{XY}(x,y)$$
$$p_Y(y) = \sum_x p_{XY}(x,y)$$

3.3-Joint and marginal probability density functions

Let X and Y be two continuous random variables with joint distribution function $F_{XY}(x, y)$, the **joint probability density function** is defined by

$$f_{XY}(x, y) := \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$$

And we can calculate

$$\iint_A f_{XY}(x, y) \mathrm{d}x \mathrm{d}y = P((X, Y) \in A)$$

Note that the probability density function satisfies that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) \mathrm{d}x \mathrm{d}y = 1$$

$Andwecandefine$f{X}(x)= \int{-\infty}^{\infty}f_{XY}(x, y) \mathrm{d}y$$ as the **marginal probability density function** (or **marginal density**) of X

3.4-Conditional distributions

In the discrete case, the **conditional probability mass function** of Y given X is

$$p_{Y|X}(y|x) := \frac{p_{XY}(x, y)}{p_X(x)}$$

And we define the **conditional probability density** of Y given $X = x$ to be

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)},$$

provided $f_X(x) \neq 0$

Theorem

Let X, Y be two random variables defined on the same probability space, then

$$E[X] = E[E[X|Y]]$$

3.5-Beyes' rule for random variables

In the case of discrete random variables X and Y,

$$p_{Y|X}(y|x) = \frac{p_{XY}(x, y)}{p_X(x)} = \frac{p_{X|Y}(x|y) p_Y(y)}{\sum_{y' \in Val(Y)} p_{X|Y}(x|y') p_Y(y')}$$

If the random variables X and Y are continuous,

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{f_{X|Y}(x|y)f_Y(y)}{\int_{-\infty}^{\infty} f_{X|Y}(x|y')f_Y(y')\mathrm{d}y'}$$

3.6-Independence of random variables

Two random variables X and Y are independent if $F_{XY}(x,y) = F_X(x)F_Y(y)$ for all values of x and y. Equivalently,

- For discrete random variables, $p_{XY}(x,y) = p_X(x)p_Y(y)$ for all $x \in Val(X), y \in Val(Y)$
- For discrete random variables, $p_{Y|X}(y|x) = p_Y(y)$ whenever $p_X(x) \neq 0$ for all $y \in Val(Y)$
- For continuous srandom variables, $f_{XY}(x,y) = f_X(x)f_Y(y)$ for all $x, y \in \mathbb{R}$
- For continuous random variables, $f_{Y|X}(y|x) = f_Y(y)$ whenever$f_X(x) \neq 0$ for all $y \in \mathbb{R}$

Lemma

If X and Y are independent then for any subsets $A, B \subseteq \mathbb{R}$, we have,

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

3.7-Expectation and covariance

Suppose that we have two random variables X, Y and $g : \mathbb{R}^2 \to \mathbb{R}$ is a function of these two random variables. Then the expectation is defined as,

- if they are discrete random variables

$$E[g(X,Y)] := \sum_{x \in Val(X)} \sum_{y \in Val(Y)} g(x,y)p_{XY}(x,y)$$

- if they are continuous random variables

$$E[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)f_{XY}(x,y)\mathrm{d}x\mathrm{d}y$$

  The covariance of two random variables X and Y is defined as

$$Cov[X,Y] := E[(X - E[X])(Y - E[Y])]$$

  And we can calculate

$$\begin{aligned} Cov[X,Y] &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY - XE[Y] - YE[X] + E[X]E[Y]] \\ &= E[XY] - E[X]E[Y] \end{aligned}$$

  When $Cov[X,Y] = 0$, we say that X and Y are uncorrelated.
  And the expectation and covariance satisfy the following properties,
- $E[f(X,Y) + g(X,Y)] = E[f(X,Y)] + E[g(X,Y)]$

- $Var[X+Y] = Var[X] + Var[Y] + 2Cov[X,Y]$
- If X and Y are independent, $Cov[X,Y] = 0$
- If X and Y are independent, $E[f(X)g(Y)] = E[f(X)]E[g(Y)]$

4-Multiple Random Variables

4.1-Basic properties

We can define the **joint distribution function**, the **joint probability density function**, the **marginal probability density function** and the **conditional probability density function** of $X_1, \ldots, X_n$, as

$$\begin{aligned}
F_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n) &= P(X_1 \le x_1, X_2 \le x_2, \ldots, X_n \le x_n) \\
f_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n) &= \frac{\partial^n F_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n)}{\partial x_1 \partial x_2 \ldots \partial x_n} \\
f_{X_1}(x_1) &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n) \mathrm{d}x_2 \ldots \mathrm{d}x_n \\
f_{X_1|X_2,\ldots,X_n}(x_1|x_2,\ldots,x_n) &= \frac{f_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n)}{f_{X_2,\ldots,X_n}(x_2,\ldots,x_n)}
\end{aligned}$$

To calculate the probability of an event $A \subseteq \mathbb{R}^n$ we have,

$$P((x_1,x_2,\cdots,x_n) \in A) = \int_{(x_1,x_2,\cdots,x_n)\in A} f_{X_1,X_2,\cdots,X_n}(x_1,x_2,\cdots,x_n) \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_n$$

Chain rule

$$\begin{aligned}
f(x_1,x_2,\cdots,x_n) &= f(x_n|x_1,\cdots,x_{n-1}) f(x_1,\cdots,x_{n-1}) \\
&= f(x_n|x_1,\cdots,x_{n-1}) f(x_{n-1}|x_1,\cdots,x_{n-2}) f(x_1,\cdots,x_{n-2}) \\
&= \cdots = f(x_1) \prod_{i=2}^{n} f(x_i|x_1,\cdots,x_{i-1})
\end{aligned}$$

Particularly, we say that random variables $X_1, \cdots, X_n$ are independent if

$$f(x_1,\cdots,x_n) = \prod_{i=1}^{n} f(x_i)$$

4.2-Random Vector, expectation and covariance

A **random vector** is a mapping $\Omega \to \mathbb{R}^n : \mathbf{X} = (X_1 \quad X_2 \quad \cdots \quad X_n)^T$

Expectation

Consider an arbitary function $g : \mathbb{R}^n \to \mathbb{R}$. The **expectation** of this function is defined as

$$E[g(X)] = \int_{\mathbb{R}^n} g(x_1,x_2,\cdots,x_n) f_{X_1,X_2,\cdots,X_n}(x_1,x_2,\cdots,x_n) \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_n$$

If g is a function from $\mathbb{R}^n$ to $\mathbb{R}^m$, i.e., if g is

$$g(x) = \begin{pmatrix} g_1(x) \\ g_2(x) \\ \vdots \\ g_m(x) \end{pmatrix}$$

Then,

$$E[g(X)] = \begin{pmatrix} E[g_1(X)] \\ E[g_2(X)] \\ \vdots \\ E[g_m(X)] \end{pmatrix}$$

Covariance

For a given random vector $X : \Omega \to \mathbb{R}^n$, its **covariance matrix** Σ is the $n \times n$ square matrix whose entries are given by $\Sigma_{ij} = Cov[X_i, X_j]$.
Then, we have

$$\begin{aligned} \Sigma &= \begin{pmatrix} Cov[X_1, X_1] & \cdots & Cov[X_1, X_n] \\ \vdots & \ddots & \vdots \\ Cov[X_n, X_1] & \cdots & Cov[X_n, X_n] \end{pmatrix} \\ &= \begin{pmatrix} E[X_1^2] - E[X_1]E[X_1] & \cdots & E[X_1X_n] - E[X_1]E[X_n] \\ \vdots & \ddots & \vdots \\ E[X_nX_1] - E[X_n]E[X_1] & \cdots & E[X_n^2] - E[X_n]E[X_n] \end{pmatrix} \\ &= E[XX^T] - E[X]E[X]^T = E[(X - E[X])(X - E[X])^T] \end{aligned}$$

Theorem

Suppose that Σ is the covariance matrix corresponding to some random vector X. Then Σ is symmetric positive semidefinite

Proof. For any vector $z \in \mathbb{R}^n$, observe that

$$\begin{aligned} z\Sigma z &= \sum_{i=1}^{n}\sum_{j=1}^{n}(\Sigma_{ij} z_i z_j) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n}(Cov[X_i, X_j] \cdot z_i z_j) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n}(E[(X_i - E[X_i])(X_j - E[X_j])] \cdot z_i z_j) \\ &= E\left[\sum_{i=1}^{n}\sum_{j=1}^{n}(X_i - E[X_i])(X_j - E[X_j]) \cdot z_i z_j\right] \end{aligned}$$

Let $x_k = X_k - E[X_k]$, Then

$$
\begin{aligned}
\sum_{i=1}^{n}\sum_{j=1}^{n}(X_i - E[X_i])(X_j - E[X_j]) \cdot z_i z_j &= \sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j z_i z_j \\
&= \sum_{i=1}^{n} x_i z_i \sum_{j=1}^{n} x_j z_j \\
&= (x^T z)^2 \\
&\geq 0
\end{aligned}
$$

And hence the expectation must be nonnegative. We conclude that $z\Sigma z \geq 0$

4.3-The law of large numbers and Central limit theorem

Theorem

Chebyshev inequality. For every random variables X that has finite variance, we have

$$P(|X - E[X]| \geq \epsilon) \leq \frac{Var[X]}{\epsilon^2}$$

Proof. Let $F_X(x)$ be the cumulative distribution function of X, then

$$
\begin{aligned}
Var[X] &= \int_{-\infty}^{\infty} (x - E[X])^2 \mathrm{d}F_X(x) \\
&\geq \int\limits_{|X-E[X]|\geq\epsilon} (x - E[X])^2 \mathrm{d}F_X(x) \\
&\geq \int\limits_{|X-E[X]|\geq\epsilon} \epsilon^2 \mathrm{d}F_X(x) \\
&= \epsilon^2 P(|X - E[X]| \geq \epsilon)
\end{aligned}
$$

So, we have

$$P(|X - E[X]| \geq \epsilon) \leq \frac{Var[X]}{\epsilon^2}$$

And equivalently,

$$P(|X - E[X]| < \epsilon) \geq 1 - \frac{Var[X]}{\epsilon^2}$$

The law of large numbers

The law of large numbers is a rule that claims the average of the results obtained from a large number of trials should converge to its expected value.
So we have the **Strong Law of Large Numbers**

Theorem

Let $X_1, X_2, \cdots$, be a series of independent and identically distributed (i.i.d.) random variables for which $E[X_1] < \infty$. Then

$$P\left(\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i = E[X_1]\right) = 1$$

The central limit theorem

Theorem

Let $X_1, X_2, \cdots$, be a series of iid random variables with mean μ and variance σ^2. Then the normalized partial sum

$$\xi_n := \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\left(\frac{X_i - \mu}{\sigma}\right)$$

satisfies

$$\lim_{n\to\infty} P(\xi_n \leq x) = \Phi(x)$$

for any x, where Φ is CDF of the standard normal distribution.